WEEK 7 – DIMENSIONALITY REDUCTION

Implementation of Feature Selection using ANOVA and PCA

Team Members

1. Neha O T - 25BAD065

2. Nitharshana S - 25BAD071

3. Rithika A M - 25BAD088

4. Shubiksha S Y- 25BAD106

5. Gopika A - 25BAD504

1. INTRODUCTION :

In this week, we worked on dimensionality reduction using the DataCo Supply Chain dataset. The main aim of this work was to identify the important features in the dataset and reduce the number of features without losing much useful information.

A dataset may contain many columns, and using all of them can make data analysis more complicated. Some features may be more useful than others. To handle this, we implemented two techniques: ANOVA for feature selection and Principal Component Analysis (PCA) for dimensionality reduction.

We also visualized the ANOVA feature scores and PCA explained variance to understand the results.

2.OBJECTIVE :

The main objectives of this work are:

- To identify important features from the dataset using the ANOVA test.

- To reduce the number of features using Principal Component Analysis.

- To understand how much information is retained after applying PCA.

- To visualize the selected features and explained variance.

- To create a final dataset containing the original features, ANOVA-selected features and PCA components.

3.DATA DESCRIPTION :

For this work, we used the DataCo Supply Chain dataset. It contains information related to orders, products, customers, sales and shipping details.

Some of the features used in our analysis include:

- Days for shipment (scheduled)
- Benefit per order
- Sales per customer
- Order Item Discount
- Order Item Quantity
- Sales
- Order Item Total
- Order Item Profit Ratio
- Product Price
- Customer Segment
- Market
- Order Region
- Shipping Mode

The target column selected for this work was Late_delivery_risk.

This target was used to identify the relationship between the selected features and delivery risk.

4.DATA PREPROCESSING :

Before applying feature selection and PCA, we prepared the dataset for analysis.

The following preprocessing steps were performed:

1. Uploaded the cleaned dataset into Google Colab.
2. Selected the required numerical and categorical columns.

3. Removed rows where the target column had missing values.

4. Converted numerical columns into numeric format.

5. Handled missing numerical values using median imputation.

6. Handled missing categorical values using most frequent value imputation.

7. Applied StandardScaler to standardize numerical features.

8. Applied One-Hot Encoding to convert categorical features into numerical format.

These steps helped us prepare the data for applying ANOVA and PCA.

5.TRAIN TEST SPLITTING:

After selecting the required features, we divided the dataset into training and testing data.

We used an 80:20 ratio.

- Training data – 80%

- Testing data – 20%

The training data was used to fit the preprocessing, ANOVA feature selection and PCA transformations. The same transformations were then applied to the testing data.

This helped us keep the testing data separate during the feature selection and dimensionality reduction process.

6. FEATURE SELECTION USING ANOVA TEST :

ANOVA stands for Analysis of Variance. It is a statistical method used to identify whether the values of a numerical feature differ across different target classes.

In our project, we used the ANOVA F-test to select important features based on the target column, Late_delivery_risk.

The F-score helps us understand how strongly a feature is associated with differences between the target classes. A higher F-score indicates a stronger difference between the class groups.

Steps followed:

1. Applied the ANOVA F-test using "f_classif".

2. Calculated the F-scores and p-values for the processed features.

3. Ranked the features based on their F-scores.

4. Selected the top 10 features using SelectKBest.

5. Displayed the selected feature names and their scores.

Result:

The ANOVA method reduced the processed feature set to the top 10 selected features. These selected features were then used as input for PCA.

Visualization:

We plotted a horizontal bar chart showing the top 10 features based on their ANOVA F-scores.

The graph helped us compare the importance scores of the selected features.

7. DIMENSIONALITY REDUCTION USING PCA

PCA stands for Principal Component Analysis. It is a dimensionality reduction technique used to convert the original features into a smaller set of new features called principal components.

These components are created by combining information from the selected features. PCA helps reduce the number of dimensions while retaining as much variation in the data as possible.

In our project, PCA was applied to the 10 features selected using ANOVA.

Steps followed:

1. Used the ANOVA-selected training features as input.

2. Applied PCA using the Scikit-learn PCA method.

3. Set the explained variance requirement to 95%.

4. Generated the required principal components.

5. Applied the same PCA transformation to the testing data.

6. Calculated the total explained variance retained.

Result:

PCA reduced the selected features into a smaller number of principal components while retaining approximately 95% of the variance in the selected training data.

This helped us represent the important information using fewer dimensions.

8. VISUALIZATION OF CUMULATIVE EXPLAINED VARIANCE :

After applying PCA, we plotted a cumulative explained variance graph.

This graph shows how the explained variance increases as the number of principal components increases.

We also included a 95% reference line to identify how many components are needed to retain at least 95% of the variance.

Result:

The graph helped us understand how many principal components were required to preserve the selected variance level.

9. 2D PCA VISUALIZATION :

We also created a two-dimensional PCA visualization using the first two principal components.

The graph displays the data points using PC1 on the X-axis and PC2 on the Y-axis. The points were coloured according to the Late_delivery_risk classes.

To make the graph easier to display, a sample of up to 3,000 training records was used.

Result:

The 2D PCA graph helped us observe how the records are distributed in a reduced two-dimensional space and understand the variation between the target classes.

10. FINAL DATASET CREATION :

After completing both ANOVA feature selection and PCA, we combined the results into a single dataset.

The final dataset contains:

1. Original selected training features.

2. Late_delivery_risk target column.

3. ANOVA-selected feature columns.

4. PCA principal component columns.

We saved the combined dataset as a CSV file using Pandas.

Output file name: "Final_Cleaned_ANOVA_PCA_Dataset.csv"

11. CONCLUSION :

In this week, we successfully implemented feature selection using the ANOVA test and dimensionality reduction using PCA on the DataCo Supply Chain dataset.

ANOVA helped us identify the top 10 important features based on their relationship with the target column, Late_delivery_risk. After that, PCA was applied to reduce the selected features while retaining approximately 95% of their variance.

We also created visualizations to understand the ANOVA scores, cumulative explained variance and two-dimensional PCA representation.

Finally, we combined the original training features, ANOVA-selected features and PCA components into a single CSV file.

Through this work, we understood how feature selection and dimensionality reduction can make a dataset easier to analyze while retaining useful information.